GeneGap | One-page hackathon write-up

An AI-assisted tool for identifying potential genetic evidence gaps in clinical trials.

Problem and user

Drug response can depend on genetic variants, but clinical trials often report demographic categories rather than participant genotypes. This makes it difficult for clinical-trial designers and pharmacogenomics researchers to assess whether trials adequately cover genetically relevant variation. GeneGap helps these users identify where evidence may be limited and which questions require further investigation.

What the system does

Input: a drug name (e.g., clopidogrel). Processing: tools retrieve known drug–gene associations and variant frequencies (CPIC/ClinVar/gnomAD) plus trial characteristics and demographics (ClinicalTrials.gov). A deterministic statistics engine estimates carrier frequencies, expected carriers, the probability of fewer than k carriers, and possible recruitment requirements where population comparisons are justified. AI agents organise the workflow and explain evidence; an independent critic verifies claims. Output: a referenced report identifying possible evidence gaps and uncertainty, or an explicit insufficient-evidence warning.

Biology and scientific approach

Clopidogrel is activated partly by the CYP2C19 enzyme. A no-function CYP2C19*2 allele can reduce activation and weaken the antiplatelet response. For a suitable autosomal biallelic variant, Hardy–Weinberg predicts the fraction carrying at least one copy as $1-(1-q)^2$, where q is allele frequency; at q=0.10 this is 19%. This is illustrative, not a measured GeneGap result. G6PD is X-linked; HLA alleles and CYP2C19 star-allele haplotypes require separate handling. Genetic ancestry is not equivalent to reported race or ethnicity.

Data and evaluation

Main sources: ClinicalTrials.gov, gnomAD, ClinVar, CPIC/PharmVar, with additional drug-target and literature sources where supported. The team has prepared a 20-pair drug–gene evidence set and a 28-trial labelling dataset; verification and provenance should be retained for each record. Planned evaluation compares GeneGap against (1) a no-tools language model and (2) demographic-only parity, using held-out reviewed records. Metrics: trial-extraction accuracy, precision/recall for drug–gene facts, unsupported-number rate and critic error-catch rate. Results are pending: no performance numbers are claimed.

Agents, risks and safeguards

Named agents retrieve targets, variants and trials, explain statistics and review evidence; maths calculations remain independent of the language model. Risks and mitigations: (1) unreliable ancestry proxies → flag/block invalid mapping and run sensitivity analyses; (2) hallucinations, source errors or prompt injection → source provenance, constrained tools and independent critic; (3) clinical overinterpretation → no individual genotype or dosing recommendations, public aggregated inputs and human sign-off.

What success looks like

A researcher can enter a drug, inspect the sources and limitations, and identify which genetic evidence gaps merit study. In the two-minute judge test, the programme should handle a new drug input, expose its research steps and either give a supported report or acknowledge insufficient information. The next practical step is a pilot with trial-design researchers.

Core references: CPIC clopidogrel–CYP2C19 guideline (2022), pmc.ncbi.nlm.nih.gov/articles/PMC9287492/ • gnomad.broadinstitute.org • clinicaltrials.gov • GeneGap Build Plan 1.